

Requirement Specification Document

Web-Based Hospital Management System

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System Topic	Web-Based Hospital Management System
Client (context)	Riverside General Hospital — a mid-sized 250-bed public hospital

1. System Purpose

The Web-Based Hospital Management System is an online platform designed to unify the core operations of Riverside General Hospital, a mid-sized 250-bed public hospital that currently runs on a mix of paper records and aging legacy software. The system replaces these fragmented tools with a single, shared patient record that every authorized department can read from and write to in real time.

Its overall aim is to bring patient records, appointment scheduling, billing and insurance, pharmacy dispensing, and laboratory orders and results together into one coherent web application accessible to patients and staff. By centralizing this information and maintaining a full audit trail behind every action, the system improves patient safety, reduces duplicate and lost records, and allows clinical and administrative staff to coordinate care efficiently across the hospital.

2. Intended Users

The system serves seven distinct user roles. Access is strictly scoped by role on a least-privilege basis, so each user sees only the slice of the shared patient record relevant to their responsibilities.

- **Patient** — Uses a self-service portal to book or cancel appointments, view their own test results once released, see and pay bills, and access their own record only.
- **Doctor** — Has full clinical read/write access for patients under their care — reviewing history, recording diagnoses, writing prescriptions, and ordering lab tests.
- **Nurse** — Views clinical data for patients on their assigned ward, records vital signs, administers and logs medication, but cannot finalize diagnoses.
- **Receptionist** — Handles patient registration, appointment scheduling, and basic billing; sees demographics and appointment status only, with no access to clinical notes.
- **Administrator** — Manages user accounts, roles, and system configuration, but has no routine access to clinical content.
- **Pharmacist** — Views active prescriptions and patient allergies, verifies and dispenses medication, and manages pharmacy inventory.
- **Lab Staff** — Receives test orders, registers specimens, and enters and releases laboratory results.

3. Functional Requirements

3.1 Patient Registration and Appointments

- Register new patients and automatically generate a unique patient ID that prevents duplicate records.
- Look up returning patients quickly by name, phone number, or patient ID.
- Book, reschedule, and cancel appointments against each doctor's live availability calendar, recording whether the visit is a first visit or a follow-up.
- Check patients in on arrival and mark them as "arrived" so the doctor's queue updates automatically.
- Send automated SMS or email appointment reminders to reduce no-shows.

3.2 Admission and Discharge

- Convert a patient from outpatient to inpatient, recording the admitting doctor, ward and bed assignment, admission date/time, and reason for admission.
- Display a live ward map showing which beds are occupied, free, or being cleaned.
- Process discharge only after the doctor authorizes it, the discharge summary is ready, take-home medication is issued, and outstanding bills are settled; the freed bed then updates to available automatically.

3.3 Clinical Care (Doctors)

- Present each doctor with a prioritized queue of scheduled appointments and assigned inpatients on login.
- Open a patient chart showing history, chronic conditions, current medications, allergies, and recent results, with critical information such as allergies displayed prominently.
- Record consultation notes, diagnoses (coded against a standard such as ICD-10), and treatment plans.
- Write prescriptions specifying drug, dosage, route, frequency, and duration, with automatic checks against patient allergies, drug-to-drug interactions, and stock availability before sign-off.
- Route a signed prescription electronically to the pharmacy queue.
- Order lab tests from within the chart, specifying the test or panel, urgency (routine or STAT), and clinical notes.

3.4 Nursing and Ward Management

- Provide a shift-handover view listing every patient on the ward, their status, outstanding tasks, and flags from the previous shift.
- Record vital signs (temperature, blood pressure, pulse, respiratory rate, oxygen saturation, pain score, blood glucose) at scheduled intervals, each stamped with the time and recording nurse.
- Log each medication dose as it is administered, building a Medication Administration Record and preventing double or missed doses.
- Maintain a shift-based task list driven by doctors' orders, with alerts for anything overdue.

3.5 Pharmacy and Inventory

- Receive signed prescriptions in the pharmacy queue, verify them, and dispense medication.
- Automatically deduct dispensed quantities from inventory and log who dispensed what, when, and to which patient.
- Maintain a drug catalogue and track stock by batch number and expiry date.
- Raise low-stock alerts at reorder thresholds, warn of medicines nearing expiry, and apply stricter logging to controlled and narcotic drugs.

3.6 Laboratory

- Receive test orders in the lab queue with patient details, the requested test or panel, urgency, and clinical notes.
- Register each specimen against its order (ideally by barcode) so a sample can never be matched to the wrong patient.
- Enter results with value, unit, and reference range, automatically flagging abnormal or critical values and alerting the ordering doctor immediately for critical results.
- Attach each released result to the correct patient and originating order, keeping a traceable link.

3.7 Billing and Patient Portal

- Generate bills for consultations, procedures, medication, and lab tests, and support both insurance and cash payment.
- Allow patients to view their records, released results, and bills, and to pay online through the self-service portal.

4. Non-Functional Requirements

- **Security** — Role-based access control, encryption of patient data at rest and in transit, an immutable audit trail on every record access, strong passwords with two-factor authentication for clinical and admin accounts, and automatic session timeouts on shared terminals.
- **Privacy and Compliance** — Compliance with health-data protection regulations, consent management for data sharing, additional access restrictions on sensitive categories, and a logged, reviewable "break-the-glass" mechanism for emergency access.
- **Availability** — 24/7 operation with a target of approximately 99.9% uptime, redundancy to avoid a single point of failure, automated daily backups, a tested disaster-recovery plan, and read-only access to critical records if the network drops.
- **Performance** — Record lookups and chart loads completing within roughly two to three seconds even at peak, and stable performance under hundreds of concurrent users.
- **Usability** — A clear, responsive web interface consistent across roles, since slow or awkward workflows lead staff to work around the system.
- **Reliability and Traceability** — Every clinical action — vitals, medication administration, dispensing, and result release — timestamped and attributed to the responsible user.

5. Constraints and Assumptions

- The system covers a single hospital with a single branch; cross-site record sharing is out of scope.
- Each patient has one unique patient ID that follows them across all visits and departments.
- A patient may be seen by many doctors over time and even within a single stay; the responsible doctor is therefore tied to each appointment and admission rather than modeled as a single permanent "primary doctor."
- Every user is assigned exactly one role, keeping the staff specialization disjoint (a person is not both, for example, a nurse and a receptionist).
- The hospital has defined departments and wards with a fixed set of beds.
- A patient is either an outpatient or an inpatient at any given time.
- The system operates in a single time zone and single currency.
- On-site internet and network connectivity is generally available, as the system is web-based.

6. Evidence of Requirement Gathering

The requirements above were gathered by interviewing five client stakeholders of Riverside General Hospital through the Project Discussion Bot: Kamal Jayasuriya (IT & Systems Administrator), Fatima Nazeer (Front Desk / Reception Supervisor), Dr. Anisha Perera (Chief Medical Officer), Nurse Manager Rohan Silva (Head of Nursing), and Dilani Fernando (Chief Pharmacist & Lab Liaison). Screenshots of these interactions, showing the questions asked and the answers received, are included below as evidence.

[Insert your six interview screenshots here — one per stakeholder, plus the scope-confirmation exchange. Add a short caption under each, e.g. "Figure 1: Interview with Kamal Jayasuriya (IT & Systems Administrator)."]